

A PROJECT TITLE
ON
Sports & Entertainment: Match Outcome Prediction Using Logistic Regression and
Random Forest

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Sports & Entertainment: Match Outcome Prediction Using Logistic Regression and Random Forest

1. ABSTRACT:

Predicting the outcome of cricket matches has become increasingly valuable to fans, broadcasters, team management, and the sports betting industry. This project aims to build a machine learning model that can accurately predict IPL (Indian Premier League) match outcomes using historical performance data, comparing an interpretable baseline against a more powerful ensemble method to identify the better-performing approach for this domain. To achieve this, we use the **IPL Complete Dataset (2008–2024)** available on Kaggle, which contains two files — matches.csv and deliveries.csv — covering match-level details (teams, venue, toss, winner, margin) and ball-by-ball data across all IPL seasons up to 2024. The dataset is pre-processed through cleaning, encoding of categorical variables, and normalization, after which two classification algorithms — Logistic Regression and Random Forest — are trained to classify each match as a win or loss for a given team.

Logistic Regression is used for its interpretability, while Random Forest is used for its ability to capture complex, non-linear feature interactions. Model performance is evaluated and compared using accuracy, precision, recall, F1 score, and confusion matrices. Experimental results show that Random Forest generally achieves higher predictive accuracy, while Logistic Regression provides clearer insight into which factors drive match outcomes — together demonstrating the practical value of machine learning in sports analytics.

2. INTRODUCTION:

Cricket, and the Indian Premier League (IPL) in particular, attracts massive global viewership, and the ability to accurately predict match outcomes carries significant value for fans, analysts, media organizations, team management, and the sports betting industry. Traditionally, match outcome prediction relied heavily on expert opinion, commentator intuition, and simple statistical indicators such as a team's win percentage or recent run rate. While useful as rough indicators, these traditional methods often fail to capture the complex, dynamic, and interacting factors — such as toss outcome, venue conditions, current squad form, head-to-head history, and player availability — that genuinely influence a game's result. As detailed ball-by-ball and match-level data has become increasingly available, machine learning has emerged as a powerful, data-driven alternative capable of uncovering patterns and relationships that go far beyond human intuition or simple heuristics.

This project addresses the problem of predicting IPL match outcomes by framing it as a binary classification task, where each match is classified as a win or loss for a chosen team (ties and no-result matches, being rare, are handled separately during pre-processing). To build and evaluate this system, we use the publicly available IPL Complete Dataset (2008–2024) from Kaggle, authored by Prateek Bhardwaj, consisting of two CSV files: `matches.csv`, which provides match-level information such as season, city, venue, teams, toss winner and decision, match winner, result margin, and player of the match; and `deliveries.csv`, which provides ball-by-ball data for every delivery bowled. The raw data is cleaned and transformed into a structured feature set — including recent team form, head-to-head statistics, venue-based (home/away) advantage, toss impact, and aggregated player/team performance indicators — suitable for supervised learning.

Two classification algorithms are applied and compared. Logistic Regression provides a simple, fast, and highly interpretable baseline that models the linear relationship between input features and the probability of a team winning, while Random Forest, an ensemble of decision trees, captures non-linear relationships and complex feature interactions with inherent resistance to overfitting. By training and evaluating both models on the same dataset, this project aims to determine which technique offers superior predictive performance, while also examining the trade-off between interpretability and predictive power. The broader goal is twofold: to develop a reasonably accurate and reliable IPL match outcome predictor using well-established, computationally lightweight techniques, and to generate insights into which factors most strongly influence match results — supporting better decision-making in team strategy, fan engagement, and sports media analytics. The findings and limitations of this approach, along with directions for future improvement, are discussed in later sections of this report.

3. SOFTWARE REQUIREMENTS:

The development and evaluation of the match outcome prediction system rely on a Python based machine learning ecosystem. The table below summarizes the software components used across data preprocessing, model building, evaluation, and reporting stages of the project.

Component	Specification / Requirement
Operating System	Windows 10 / 11, Linux (Ubuntu 20.04+), or macOS
Programming Language	Python 3.9 or above
IDE / Notebook Environment	Jupyter Notebook, Google Colab, VS Code, or PyCharm
Core Libraries	NumPy, Pandas (data handling and preprocessing)
Machine Learning Libraries	Scikit-learn (Logistic Regression, Random Forest, model evaluation)
Visualization Libraries	Matplotlib, Seaborn (exploratory data analysis and result plots)
Data Source / Storage	CSV / Excel datasets, or SQL database for historical match data
Version Control	Git and GitHub for source code management
Documentation Tools	Microsoft Word / LaTeX for report preparation
Web Framework (optional)	Flask / Streamlit for deploying the prediction model as a demo application

4. HARDWARE REQUIREMENTS:

Since Logistic Regression and Random Forest are computationally lightweight compared to deep learning models, the project does not require specialized high performance hardware. A standard personal computer or laptop meeting the following specifications is sufficient for data preprocessing, model training, and evaluation.

Component	Specification / Requirement
Processor	Intel Core i5 (8th Gen) / AMD Ryzen 5 or higher
RAM	8 GB minimum (16 GB recommended for faster model training)
Storage	256 GB SSD or higher, with at least 5 GB free space for datasets and libraries
Graphics	Integrated graphics sufficient (dedicated GPU not mandatory for these algorithms)
Internet Connectivity	Required for downloading datasets, libraries, and using cloud notebooks
Display	Standard monitor with minimum 1366 x 768 resolution
Input Devices	Keyboard and mouse

Signature of the Guide

Course Instructor

